```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow

# Upload image
uploaded = files.upload()

# Get image name
filename = list(uploaded.keys())[0]

# Read image
image = cv2.imread(filename)

# Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Calculate gradient along X-axis
gradient_x = cv2.Sobel(
    gray,
    cv2.CV_64F,
    1,
    0,
    ksize=3
)

# Calculate gradient along Y-axis
gradient_y = cv2.Sobel(
    gray,
    cv2.CV_64F,
    0,
    1,
    ksize=3
)

# Calculate gradient
gradient = cv2.subtract(gradient_x, gradient_y)

# Convert to 8-bit image
gradient = cv2.convertScaleAbs(gradient)

# Save output
cv2.imwrite("sharpened_image3.jpg", gradient)

# Display original image
print("Original Image:")
cv2_imshow(image)

# Display gradient masked image
print("Gradient Masked Image:")
cv2_imshow(gradient)
```

```
print("Output saved as sharpened_image3.jpg")
```

Choose files img3.jpeg

img3.jpeg(image/jpeg) - 10583 bytes, last modified: 20/07/2026 - 100% done

```
Saving img3.jpeg to img3 (1).jpeg
Original Image:
```


```
Gradient Masked Image:
```

```
Output saved as sharpened_image3.jpg
```